# NGUYEN MINH DUC


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- Ho Chi Minh City


## Skills

- **IP/SoC Design**
- **ASIC/FPGA Design Flow**
- **UVM Testbench**
- **RISC-V Architecture**
- **Cadence Tools**
- **Python/Scripts/Tcl/Perl**
- **Verilog/SystemVerilog**
- **C/C++/Java**
- **English Communication**

## Education

**International University**

__ Sep. 2021 - Sep. 2025

Bachelor of Engineering of Information Technology

**University of Technology**

__ Jan. 2026 - Jan. 2028

Master of IC Design

100% CT Group Scholarship

## Achievements

**Overall GPA 3.48**

**Learning Scholarship 2021-2022 & 2024-2025**

**IELTS 7.0**

## Link

Personal Profile

Project for Study

Individual Project

## Professional Summary

First-year master's program of IC Design with a 100% tuition fee scholarship with hands-on experience as a teaching assistant at an IC design training center, seeking a front-end IC Design Engineer role to leverage strong technical and problem-solving skills. Solid knowledge in computer architecture and IP/SoC design principles. Proactively pursues professional development by mastering new concepts such as AI, machine learning, etc., integrating constructive suggestions, and responding positively to feedback. Familiar with HDLs like Verilog/SystemVerilog and other coding languages such as Python/Tcl/Perl/Scripts. Strong project time management and communication skills by consistently meeting deadlines and facilitating effective cross-team collaboration.

## Projects

**University Projects and Labs** September 2021 - September 2025

- **Year 1** - Learning general subjects and solving coding quiz in C/C++.
- **Year 2** - Studying pseudocode and algorithms in Computer Architecture major, coding game project in groups of 4-5 students in OOP, DSA majors, working with lab machines in Digital & Signal majors using Java, C/C++, Python.
- **Year 3** - Orienting to Computer Engineering specialization. Deeply learning and researching into special fields associating with IC design, implementing logic gates and small projects on FPGA kit using HDL (Verilog/VHDL) parallel with scripting language (C/C++, Python), studying virtual hardware (Ubuntu Linux).
- **Year 4** - Joining internship at SEMICON Center, researching for prethesis (***"Design a single cycle RISC-V CPU Core on FPGA"***) and thesis as final project following the guidance of university's mentor. The thesis title is ***"Implementation of machine learning for MNIST Digits Recognition on FPGA"***. Learning additional trending subjects like AI, and VLSI Concepts.

**Individual Projects** __ March 2025 - Now

- ***SEMICON Basic Course Practices*** - Learning about basic design and verification of small components based on given specs using Verilog and Shell script. Study deeply the ASIC design flow from creating design spec to simple simulation.
- ***Cadence Virtuoso*** - Designing simple schematic and layout of logic gates. Calculating simple timing delay and checking errors by running LVS, DRC, etc.
- ***UVM*** - Design testbench for small projects based on UVM principles using SystemVerilog.
- ***Perl/Tcl/Shell*** - Studying syntax and application into real projects.

## Work History

**Teaching Assistant** __ May 2025 - Dec 2025

Microchip Design Training Center SEMICON -

Thuy Loi 4 Building, 205 Nguyen Xi, Binh Thanh Ward, Ho Chi Minh City

- Support new members at offline and online basic course classes.
- Provide practice lessons guidance to all the members.
- Correct homework and resolve technical issues.